

Mediated model collapse: empirical findings

(companion to `theory_contamination.tex`, Parts A–E)

This note reports the measurements and pairs each with the theory part it confirms, refines, or bounds. The theory document is the proven framework; here we say what the experiments show, and we are explicit about what is proven, what is measured, and what is only suggestive. Tiers: an *MLP toy* (fast, with exact provenance tags) and the *LLM tier* (Qwen2.5-7B, LoRA SFT-KL, real Pokec graph and profiles).

1 The loop

Each round: a nonlinear bounded-confidence population (Hegselmann–Krause, or Deffuant with algorithmic bias) updates; a model is retrained on the current (platform-mediated) opinions, anchored to a fixed prior with strength β (the SFT-KL coefficient); the model feeds a *per-node* prediction μ_i back, absorbed only by agents within the confidence radius ε . Contamination s is the fraction of opinion that is recycled model output.

2 Contamination follows the identity (Parts A, C)

MLP, exact. The gated identity $\Delta s_t = w \text{contact}_t \langle 1-f \rangle_{G_t}$ holds to machine precision; the measured law $ds/dt \propto \text{contact}(1-s)$ is its uncorrelated case; forced detachment stalls s at the captured fraction ($s \approx 0.69$), as the covariance correction predicts. The degeneracy (retraining does not move s) is verified symbolically and numerically. These are the proven core (Part A) made visible.

3 The β switch: displacement vs collapse (Part B)

LLM, two seeds — the headline. The KL anchor selects the failure mode:

- $\beta = 0$: population diversity *contracts* (std 0.094 $\rightarrow$ 0.067 at $\varepsilon = 0.4$) with small mean displacement (+0.05) — **collapse**;
- $\beta = 3$: diversity is *maintained/widened* (std 0.114 $\rightarrow$ 0.153) with large displacement (+0.17 toward the prior) — **displacement without collapse**.

This is the gated, nonlinear realization of the *mechanism* in Part B’s variance decomposition $\text{Var}(x_\infty) = (1-s)^2 \text{Var}(x_0) + s^2 \text{Var}(\boldsymbol{\mu}) + 2s(1-s) \text{Cov}$ (which is itself a linear-model identity): collapse is the concentrated-prediction case ($\text{Var}(\boldsymbol{\mu}) \rightarrow 0$, $\beta = 0$), displacement the spread-prediction case ($\text{Var}(\boldsymbol{\mu}) > 0$, $\beta = 3$). The qualitative driver — whether the model’s predictions are uniform or spread — transfers from the linear formula to the gated experiment; the closed form itself does not (the gated dynamics also carry the floor of Part E). The per-node structure is load-bearing either way: a scalar-broadcast model could only ever collapse.

4 Gate-monotone, loop-caused collapse (Parts C, D)

LLM, two seeds. Model-side collapse is measured by ℓ_{c0} , the current model’s loss on the frozen round-0 data (forgetting of the original distribution). Peak ℓ_{c0} rises monotonically with the gate:

no-feedback 1.25 < $\varepsilon=.1$ 1.85 < .2 2.59 < .3 3.36 < .4 3.98.

The flat no-feedback control (ℓ_{c0} barely moves) proves the loop causes the forgetting; the trust radius ε sets the depth. This is the model-side shadow of Part D’s capture region: wider ε captures more of the population, so the model trains on more-collapsed data. Reproduced across both seeds; the $\varepsilon=.3$ vs .4 ordering, unsettled at 30 rounds, resolves ($\varepsilon=.4$ deeper) at 60.

5 The diversity floor (Part E)

Captured populations do not reach zero spread: the residual std is 0.07–0.10, matching the floor $A^2\sigma_m^2$ with σ_m the model’s per-node prediction spread. Diversity reaches zero only when predictions concentrate ($\sigma_m \rightarrow 0$), tying back to the β switch.

6 The anchor does double duty

β does not only prevent collapse — it also *stabilizes training*. At $\beta = 0$, continual SFT destabilizes at long horizons (around round 50, the loss on the current batch jumps as the model stops fitting its own data); every $\beta > 0$ run stays stable. The clean collapse metric is therefore the *peak* ℓ_{c0} , not the final value.

7 Collapse depth: the fraction-saturated corner (P5, honest)

The natural prediction $\ell_{c0} \sim s^2$ **fails**, and instructively. Contamination saturates ($s \rightarrow 1$ in every loop cell), so the synthetic *fraction* is pinned and cannot explain the ε -ladder. The depth instead tracks the training-data *quality*: ℓ_{c0} rises as the batch variance falls ($0.016 \rightarrow 0.005$ as ℓ_{c0} 1.85 $\rightarrow$ 3.98). The decisive pair: no-feedback and $\varepsilon=.2$ share the same batch variance but differ in ℓ_{c0} (1.25 vs 2.59) — because one trains on real data ($s = 0$) and the other on recycled data ($s \approx 1$). So in the two-factor law $\ell_{c0} = \text{base} + s^2 c^2$, the fraction s^2 is a *switch* (real vs recycled) and the quality gap c^2 carries all the ε -dependence. Our loop retrain on the fully-mediated population every round, so it sits in the fraction-saturated, quality-varying corner — distinct from the fraction-dialing regime of the model-collapse literature. The bridge holds structurally; the precise c^2 -vs-quality form awaits more seeds (and is muddled by the $\beta = 0$ instability).

8 Real-data validation

Pokec (unimodal innate, sparse graph): the ε axis survives — threshold, gated capture, stranded minority — while the algorithmic-bias (γ) axis is inert (the graph is too sparse and the population too unimodal to fragment). *PolitiSky24* (real polarized 2024-election stances on a dense interaction graph): γ matters; a centrist-prior platform merges both camps at its prior; a partisan-prior platform flips the majority camp at wide ε and strands roughly half of it at narrow ε . Caveats: PolitiSky labels are LLM-annotated and the camps are 13:1 imbalanced.

9 Limitations

Most cells are one or two seeds. The $\beta = 0$ long-horizon instability corrupts the collapse tail (we report the peak). The inherited platform susceptibility is high, so magnitudes are conditional on strong trust. The γ -healing result needs a denser real graph to test fairly. LLM contamination s is reconstructed from the contact trajectory via the Part-A identity, not directly tagged; a provenance tag in the pipeline would let us measure it.

10 Status

Proven and verified (Parts A–E): contamination is value-blind with closed-form s and a degeneracy; per-node displacement and the three-term variance; the gated s -identity; the capture/detachment boundary ε/κ ; the diversity floor $A^2\sigma_m^2$. Measured and reproduced (two seeds): the β switch, gate-monotone loop-caused collapse, the floor magnitude, the anchor’s stabilizing role. Refined, not closed: the collapse-depth law (P5, fraction-saturated corner). Conjectural: the performative-stable-point framing (P6), with the caveat that the closed-form KL result is RL-KL while our experiments are SFT-KL.